Retrieval augmented generation

Retrieval augmented generation, or RAG, retrieves relevant passages from a document collection and supplies them to a language model to help generate a grounded answer.

Embeddings and semantic search

An embedding represents text as a numerical vector. Semantic search compares vector representations to find passages that are conceptually related, even when wording differs.

Limitations of RAG

RAG quality depends on document coverage, chunk boundaries, retrieval ranking, and the language model's use of retrieved evidence. Retrieved context does not guarantee a correct answer.